

PolypDG-Lite: A Deployment-Oriented Approach for Cross-Center Colonoscopic Polyp Segmentation via Domain Generalization and Low-Power Knowledge Distillation

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Abstract

Robust colonoscopic polyp segmentation under cross-center domain shift remains a major challenge for practical computer-aided endoscopy. Models trained on data from limited institutions often suffer from performance instability when evaluated on unseen centers. This is primarily due to variations in imaging pipelines, lighting, device-specific color responses, compression, and operator-dependent acquisition styles. This study evaluates cross-center robustness using the multi-center PolypGen dataset and a strict leave-one-center-out (LOCO) protocol, where each test fold consists of data from a strictly unseen institution. Our proposed PolypDG-Lite is a deployment-oriented framework that integrates domain generalization, teacher-student knowledge distillation, and low-power inference validation. Moving beyond simple overlap-based metrics, this work introduces a risk-oriented evaluation framework that accounts for worst-case behavior, cross-center performance gaps, boundary-sensitive distances, and deployment costs. Among the evaluated SegFormer-B2 teacher variants, consistency learning achieves the most reliable cross-center performance. Compared to the U-Net baseline, the mean Dice score increased from 63.40% to 79.61%. On the most challenging unseen center (C4), the Dice score increased from 43.45% to 66.13%, demonstrating superior worst-center robustness. To facilitate lightweight deployment, the trained teacher model is distilled into multiple compact students, including DDRNet-23-slim, BiSeNetV2, and SegFormer-B0. Integrating knowledge distillation with SegFormer-B0 yields the optimal balance between robustness and efficiency, achieving a mean Dice score of 77.05% and a C4 Dice score of 60.44%. Utilizing FP16 inference, the final student model maintains its full LOCO performance with a compact 7.08 MB footprint, while cutting peak GPU memory to 55.43 MB and achieving 95.26 FPS. Furthermore, low-power analysis reveals a mean power consumption of 14.77 W and an energy efficiency of 0.3818 J/frame. These findings indicate that PolypDG-Lite mitigates cross-center degradation while enabling lightweight, low-power deployment for polyp segmentation.

Keywords: Colonoscopic Polyp Segmentation, Cross-Center Generalization, Domain Shift, Leave-One-Center-Out (LOCO), Knowledge Distillation, Low-Power Deployment, PolypGen

1. Introduction

Colonoscopy is a critical clinical procedure for detecting and removing colorectal polyps, a preventive measure essential to reducing colorectal cancer incidence. In computer-aided endoscopy, colonoscopic polyp segmentation aims to generate pixel-level lesion masks from endoscopic images, providing precise information regarding polyp location, boundaries, and spatial extent. Compared to image-level classification or bounding-box detection, segmentation offers a finer-grained representation of lesion morphology, thereby enhancing downstream clinical interpretation, quantitative assessment, and decision support. However, reliable polyp segmentation remains challenging due to substantial variations in polyp size, shape, color, and texture, further compounded by illumination changes, specular highlights, blur, and various imaging artifacts.

Furthermore, the boundary between a polyp and the surrounding mucosa is often indistinct, compromising accurate and stable mask prediction in practical clinical scenarios.

Deep learning has significantly advanced medical image segmentation. Ronneberger et al. [1] established a widely adopted baseline with U-Net, introducing an encoder–decoder architecture with skip connections. For colonoscopic polyp segmentation, Fan et al. [2] developed PraNet to refine polyp regions using a reverse attention mechanism, while efficient designs like HarDNet-MSEG [3] demonstrated the feasibility of balancing segmentation accuracy with inference speed. More recently, SegFormer [4] integrated a hierarchical Transformer encoder with a lightweight decoder for robust semantic segmentation. Although these methods continuously push performance boundaries, high average Dice or IoU scores on standard benchmarks do not guarantee reliable cross-center deployment. Indeed, average scores can obscure severe performance degradation on challenging centers, unstable boundary predictions, small-polyp omissions, and false-positive responses.

Ali et al. [5] introduced PolypGen as a multi-center benchmark to evaluate polyp detection and segmentation model generalization across clinical centers. Real-world deployments face center-specific domain shifts caused by variations in imaging hardware, illumination, color response, acquisition, and compression. Conventional evaluations using randomly split datasets can therefore overestimate practical reliability. To mitigate this, we employ a center-level leave-one-center-out (LOCO) validation protocol, where each center serves as an unseen test domain. Beyond standard mean Dice and IoU scores, our evaluation prioritizes worst-center performance, validation-to-test performance degradation, and boundary-sensitive metrics (HD95 and ASSD) for a risk-aware assessment of cross-center robustness.

Another practical challenge is deployment efficiency. While high-capacity models achieve strong segmentation accuracy, they often demand substantial memory, high inference latency, and significant power consumption. Conversely, lightweight models are better suited for real-time or resource-constrained deployment but often suffer from limited representation capacity and reduced robustness under domain shift. Knowledge distillation (KD) addresses this trade-off by transferring the knowledge of a powerful teacher model to a compact student model [6]. For deployment-oriented segmentation, lightweight architectures like DDRNet [7] and BiSeNetV2 [8] highlight the critical need to balance segmentation accuracy with computational efficiency.

In this paper, we propose PolypDG-Lite, a deployment-oriented framework for cross-center colonoscopic polyp segmentation. Using the clean, center-labeled subset of PolypGen, the framework establishes a rigorous center-level LOCO evaluation protocol. We first employ a U-Net baseline to diagnose cross-center performance degradation under unseen-center conditions. Next, a SegFormer-B2 teacher is trained using robustness-oriented consistency learning, and knowledge distillation transfers its segmentation capabilities to lightweight student models, including DDRNet-23-slim, BiSeNetV2, and SegFormer-B0. Finally, the selected student model is evaluated across multiple deployment metrics: FP16 inference speed, runtime latency, GPU memory usage, power consumption, and energy-per-frame consumption. Ultimately, this study aims not only to improve average segmentation accuracy but also to establish a comprehensive evidence chain spanning cross-center robustness evaluation, lightweight knowledge distillation, and low-power deployment validation.

2. Methodology

PolypDG-Lite is designed to address two key challenges in colonoscopic polyp segmentation: cross-center domain shift and deployment efficiency. Figure 1 outlines the overall workflow. First, a clean, center-labeled subset from the PolypGen dataset [5] is constructed, and a center-level LOCO protocol is applied to assess unseen-center generalization. Guided by this protocol, the framework executes four sequential stages: U-Net baseline establishment, robustness-oriented SegFormer-B2 teacher training, lightweight student knowledge distillation, and low-power FP16 deployment profiling.

Unlike evaluations based on randomly mixed train-test splits, this protocol partitions data at the clinical-center level. Consequently, each test fold represents a strictly unseen institution, providing a realistic assessment of cross-center robustness

under domain shift [9]. To guarantee a fair comparison across the baseline, teacher, student, and deployment settings, we maintain identical data subsets, LOCO splits, preprocessing procedures, and evaluation metrics throughout all experimental stages.

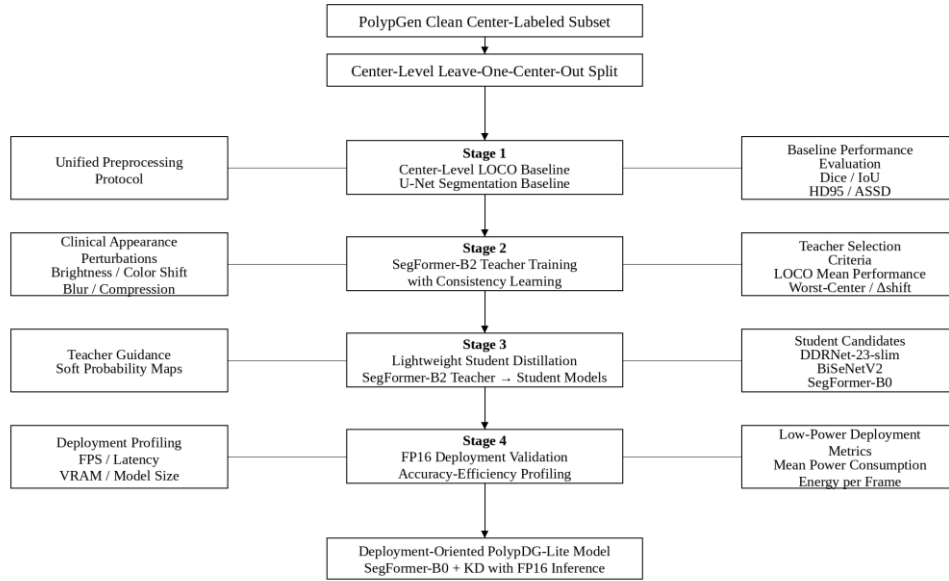


Fig. 1 Overall Workflow of the Proposed PolypDG-Lite Framework

2.1. Dataset Construction and Center-Level LOCO Split

This dataset construction and center-level splitting strategy form the foundation of the PolypDG-Lite framework. Because this study focuses on cross-center generalization, the dataset is partitioned by clinical site rather than divided into random train-test subsets. Random splitting often allows images from the same center to appear in both training and testing sets, leading to an overly optimistic estimation of model performance under clinical domain shift [9]. Conversely, center-level separation rigorously evaluates whether a model can generalize to an institution completely unseen during training.

The original PolypGen dataset contains 8,037 image-mask pairs designed for multi-center polyp detection and segmentation evaluation [5]. For the formal LOCO experiments in this study, we utilize a clean, center-labeled subset. This subset comprises 1,537 image-mask pairs with explicit source labels across six centers, denoted as C1 to C6. The image counts for C1 through C6 are 256, 301, 457, 227, 208, and 88, respectively. This fixed subset is maintained throughout all experimental stages to guarantee that performance variations stem strictly from architectural design rather than data selection discrepancies.

Under the center-level LOCO protocol, each fold reserves one center as an unseen test domain while utilizing the remaining five centers for training and validation. To prevent data leakage, the validation set is sampled exclusively from the training centers, ensuring no images from the held-out center are present during model development. Each center serves as the unseen test domain exactly once. This setup enables comprehensive center-wise performance analysis, worst-center identification, and the measurement of validation-to-test degradation under cross-center domain shift.

2.2. Unified Preprocessing and Evaluation Metrics

To ensure a fair comparison across all stages, a unified preprocessing pipeline is applied to all experiments. Input images and segmentation masks are resized to a fixed spatial resolution prior to model training and evaluation. Specifically, image resizing is performed using bilinear interpolation, whereas mask resizing utilizes nearest-neighbor interpolation to preserve discrete label values. Finally, image pixels are normalized to the range [0, 1], and segmentation masks are binarized into

foreground polyp and background regions.

Data augmentation is applied exclusively to the training set. The operations include random flipping, rotation, scaling, and brightness–contrast adjustments, which enhance model exposure to common appearance variations in colonoscopic images. Conversely, validation and test images undergo only deterministic resizing and normalization without random augmentations to ensure reproducible and stable evaluation.

Model performance is evaluated using both region-level and boundary-level metrics. The Dice similarity coefficient and Intersection over Union (IoU) serve as the primary overlap-based measures, while accuracy, precision, and recall provide complementary pixel-level insights. The Dice and IoU metrics are defined as follows:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} \quad (1)$$

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (2)$$

where TP, FP, and FN denote true-positive, false-positive, and false-negative pixels, respectively.

While Dice and IoU measure region-level overlap, they may not fully capture boundary displacement or contour instability. Therefore, we incorporate HD95 (95th percentile Hausdorff Distance) and ASSD (Average Symmetric Surface Distance) as boundary-sensitive metrics to quantify spatial boundary errors in segmentation evaluation [10]. Beyond center-wise results, this study reports the overall LOCO mean, worst-center performance, and validation-to-test degradation. This degradation measures the performance drop from the validation centers to the unseen test center, formally defined as:

$$\Delta\text{Dice} = \text{Dice}_{\text{val}} - \text{Dice}_{\text{test}} \quad (3)$$

$$\Delta\text{IoU} = \text{IoU}_{\text{val}} - \text{IoU}_{\text{test}} \quad (4)$$

A positive value indicates performance degradation from the validation centers to the unseen test center, whereas a negative value signifies that the held-out test center outperforms the validation baseline in that fold.

2.3. Stage 1: U-Net Baseline Under LOCO Protocol

Stage 1 establishes a U-Net baseline under the center-level LOCO protocol. U-Net is selected because it represents a foundational encoder–decoder architecture in biomedical image segmentation, providing a classic benchmark for performance evaluation [1]. Rather than serving as our final proposed model, U-Net is utilized here as a diagnostic baseline to analyze how a standard medical segmentation model behaves under strict unseen-center conditions.

For each LOCO fold, the U-Net model is trained on the designated training centers, with hyperparameter tuning and model selection guided strictly by validation performance within those same centers. The held-out center is reserved exclusively for final testing and remains completely decoupled from training, validation, and model selection. This setup guarantees that the baseline results reflect true unseen-center generalization rather than performance inflated by randomly mixed data.

The output of Stage 1 is a center-level baseline profile comprising per-center segmentation metrics, overall LOCO mean performance, worst-center performance, boundary-sensitive errors, and validation-to-test shifts. These results establish the reference baseline for subsequent teacher training, student distillation, and deployment validation stages. More importantly, Stage 1 serves to diagnose whether cross-center domain shift induces uneven performance degradation across different clinical

centers.

2.4. Stage 2: SegFormer-B2 Teacher Training with Consistency Learning

Stage 2 trains a SegFormer-B2 teacher model to enhance cross-center segmentation robustness beyond the U-Net baseline. SegFormer is selected because its hierarchical Transformer encoder provides robust semantic representation, while its lightweight decoder enables efficient dense prediction [4]. Compared to the U-Net baseline, SegFormer-B2 is expected to deliver superior feature abstraction and more stable segmentation performance under center-level variations.

This stage evaluates four teacher configurations under the identical LOCO protocol and preprocessing pipeline. The first configuration serves as the SegFormer-B2 baseline without additional robustness strategies. The second configuration introduces a clinical perturbation bank that applies appearance-level transformations—such as brightness changes, color shifts, blur, and compression artifacts—to approximate clinically observable variations rather than arbitrary distortions. The third configuration implements consistency learning, regularizing the model to produce stable segmentation predictions across different perturbed views of the same input. This approach is motivated by consistency regularization, a well-established strategy to improve model stability under input perturbations [11], [12]. Finally, the fourth configuration combines both the perturbation bank and consistency learning to investigate whether their mechanisms offer complementary robustness.

Under the consistency learning framework, two appearance-perturbed views are generated for each input image while maintaining strict spatial alignment with their shared segmentation mask. The model is then optimized using a supervised segmentation loss coupled with a consistency objective enforced between the prediction probability maps of the two views. The total teacher training objective is formulated as:

$$\mathcal{L}_{teacher} = \mathcal{L}_{seg} + \alpha \mathcal{L}_{con} \quad (5)$$

where $\mathcal{L}_{seg}$ denotes the supervised segmentation loss, $\mathcal{L}_{con}$ denotes the consistency loss between the prediction probability maps of two perturbed views, and α controls the contribution of the consistency term.

This design forces the teacher model to rely less on center-specific appearance cues and focus more on domain-invariant polyp structures. Under the LOCO protocol, this prediction stability is expected to enhance robustness on unseen centers. Stage 2 yields fold-specific teacher checkpoints alongside comprehensive LOCO evaluation profiles. Crucially, teacher selection is not based solely on mean Dice or IoU scores; instead, the optimal configuration is determined by jointly considering the LOCO mean, worst-center performance, validation-to-test degradation, HD95, and ASSD. Once selected, this SegFormer-B2 teacher model is fixed to serve as the exclusive knowledge source for Stage 3 knowledge distillation.

2.5. Stage 3: Lightweight Knowledge Distillation

Stage 3 applies KD to transfer the robust segmentation capabilities of the selected SegFormer-B2 teacher to lightweight student models. While high-capacity teachers deliver superior performance, their substantial memory footprint, latency, and computational cost hinder real-world deployment. Conversely, lightweight models lower deployment overhead but often sacrifice robustness under cross-center domain shift. To address this trade-off, this stage evaluates whether teacher-guided supervision can effectively help compact architectures preserve segmentation performance across unseen domains [6, 13].

During distillation, the selected SegFormer-B2 teacher is frozen, and its weights receive no further updates. For each input image, the teacher generates a pixel-wise soft probability map. Compared to using the binary ground-truth mask alone, this soft probability map provides richer spatial supervision, particularly around ambiguous boundary regions and foreground-background transitions.

This guidance is particularly beneficial for dense prediction tasks, where the spatial structure of teacher predictions provides crucial contextual cues for student networks [13]. Accordingly, each student model is optimized using a joint loss function comprising the supervised segmentation loss and the distillation loss. The total student optimization objective is formulated as:

$$\mathcal{L}_{student} = \mathcal{L}_{seg} + \lambda \mathcal{L}_{KD} \quad (6)$$

where $\mathcal{L}_{seg}$ denotes the supervised segmentation loss between the student prediction and the ground-truth mask, $\mathcal{L}_{KD}$ denotes the distillation loss between the student probability map and the teacher probability map, and λ controls the contribution of the distillation term. The distillation term can be represented as:

$$\mathcal{L}_{KD} = D(P_s, P_t) \quad (7)$$

where P_s and P_t denote the student and teacher probability maps, respectively, and $D(\cdot)$ denotes the distance or divergence function used to align the two probability distributions.

Three lightweight student candidates are evaluated: DDRNet-23-slim, BiSeNetV2, and SegFormer-B0. DDRNet-23-slim is included as an efficient dual-resolution segmentation model [7]; BiSeNetV2 is selected as a real-time bilateral segmentation architecture [8]; and SegFormer-B0 is used as a compact Transformer-based student from the SegFormer family [4]. These candidate networks represent distinct lightweight designs, enabling us to evaluate whether knowledge distillation performance varies across different student architectures.

Stage 3 yields a set of distilled lightweight student models evaluated under the identical center-level LOCO protocol. Rather than relying solely on average Dice and IoU scores, model selection jointly considers worst-center performance, boundary-sensitive metrics, and validation-to-test degradation. The optimal student network is ultimately designated as the deployment candidate for Stage 4.

2.6. Stage 4: FP16 Deployment Validation and Low-Power Profiling

Stage 4 evaluates the deployment behavior of the selected lightweight student model. This stage involves no retraining; instead, it benchmarks the frozen checkpoints under production-oriented inference settings to determine whether the chosen model maintains segmentation quality while reducing computational overhead.

First, an FP16 LOCO evaluation is conducted to examine whether reduced-precision inference preserves segmentation performance across unseen centers. FP16 inference is adopted because half-precision computation reduces memory usage and improves inference throughput on compatible hardware [14-15]. This step maintains the identical center-level LOCO protocol and evaluation metrics—including Dice, IoU, accuracy, precision, recall, HD95, and ASSD. Crucially, the validation-to-test shift is monitored to verify whether FP16 inference introduces additional cross-center instability.

Second, we measure runtime and memory efficiency using key deployment metrics: frames per second (FPS), mean latency, latency jitter, peak GPU memory usage, parameter count, and checkpoint size. These metrics are evaluated under a uniform benchmarking configuration to contrast the teacher model, the lightweight student baseline, the distilled student, and the FP16 distilled student.

Finally, low-power profiling is performed to evaluate energy efficiency during inference. Alongside mean and peak power consumption, we measure dynamic power and energy-per-frame consumption. The energy per frame is formally defined as:

$$E_{\text{frame}} = \frac{P_{\text{mean}}}{\text{FPS}} \quad (8)$$

where E_{frame} denotes the energy consumption per processed frame, P_{mean} represents the mean power consumption during inference, and FPS signifies the inference throughput in frames per second.

FPS alone is insufficient for deployment evaluation because a model with high throughput may still exhibit excessive power consumption. Therefore, energy per frame is included to quantify the computational energy required to process each image. Ultimately, the final PolypDG-Lite model is selected based on a balanced trade-off among segmentation quality, cross-center robustness, runtime efficiency, memory footprint, and low-power inference behavior.

3. U-Net Baseline Evaluation under Center-Level LOCO

Stage 1 evaluates the U-Net baseline under the center-level LOCO protocol to analyze the effects of cross-center domain shift in colonoscopic polyp segmentation. U-Net is adopted as a representative biomedical segmentation baseline due to its foundational encoder-decoder structure and skip connections, which are widely utilized in medical image segmentation [1]. The purpose of this stage is not to propose U-Net as the final model, but rather to establish a diagnostic baseline to understand how standard medical segmentation networks behave when evaluated on strictly unseen clinical centers.

To evaluate how cross-center domain shift affects generalization across institutions, our analysis reports center-wise overlap metrics, boundary-sensitive errors, and validation-to-test gaps. These indicators characterize region-level accuracy, boundary reliability, and cross-center robustness, respectively [5, 9]. Crucially, the baseline evidence established in this section underpins the subsequent SegFormer-B2 teacher training and lightweight knowledge distillation stages.

3.1. Center-Level Baseline Performance

Table 1 presents the center-level performance of the U-Net baseline under the LOCO protocol. Across the six held-out centers, the baseline obtains a mean Dice of 63.40% and a mean IoU of 55.07%. Although these averages provide an overall summary of segmentation performance, the center-wise results reveal substantial variability across unseen domains. While C3 achieves the highest Dice score of 78.68%, C4 drops to the lowest at 43.45%, resulting in a 35.23 percentage-point gap between the best- and worst-performing centers. This substantial performance disparity demonstrates that the baseline fails to generalize uniformly across different clinical environments.

Table 1. Center-level performance of the U-Net baseline under the center-level LOCO protocol

Center	Num_images	Dice	IoU	Accuracy	Precision	Recall	HD95	ASSD
C1	256	72.14%	62.85%	96.02%	77.92%	73.40%	60.17	19.56
C2	301	61.21%	53.08%	94.56%	66.26%	63.63%	82.48	29.98
C3	457	78.68%	69.54%	96.88%	85.35%	78.01%	48.20	14.44
C4	227	43.45%	36.98%	93.69%	46.44%	50.92%	90.54	35.39
C5	208	52.72%	43.98%	93.65%	65.59%	56.43%	82.90	31.58
C6	88	72.21%	64.00%	95.71%	77.31%	74.56%	62.47	22.85

These findings indicate that the impact of cross-center domain shift is highly center-dependent. Specifically, the model achieves Dice scores above 72% on C1, C3, and C6, demonstrating relatively stable region-level segmentation on these held-

out domains. Conversely, performance drops sharply on C4 and C5, yielding Dice scores of 43.45% and 52.72%, respectively. This pattern reveals that the U-Net baseline remains highly sensitive to center-specific visual attributes, including imaging appearance, illumination, color response, and acquisition style. Such localized degradation underscores the necessity of multi-center generalization assessments in colonoscopic polyp segmentation [5].

Boundary-sensitive metrics further confirm the instability observed across challenging centers. Specifically, C4 exhibits the largest boundary error, with an HD95 of 90.54 and an ASSD of 35.39, closely followed by C5 with an HD95 of 82.90 and an ASSD of 31.58. Because overlap metrics alone often fail to capture spatial contour discrepancies, HD95 and ASSD are essential for evaluating boundary-level reliability [10]. These findings indicate that performance drops on difficult domains are not limited to reduced region overlap but heavily involve unstable boundary delineation. Clinically, such boundary errors can compromise the reliability of lesion extent estimation even when partial overlap is maintained.

Overall, the Stage 1 baseline exposes a clear center-wise generalization bottleneck under the LOCO protocol. Relying solely on mean Dice and IoU scores is insufficient to guarantee cross-center reliability, as these averages can mask severe performance degradation on specific unseen centers. Consequently, worst-center performance and boundary-sensitive metrics are indispensable for evaluating whether a segmentation model remains reliable under clinical domain shift [9-10]. This critical observation directly motivates the robustness-oriented teacher training evaluated in the subsequent stage.

3.2. Domain Generalization Gap under LOCO

To examine whether validation metrics accurately reflect model behavior on unseen centers, Table 2 reports the domain generalization gaps of the U-Net baseline under the LOCO protocol. These gaps are computed using the validation-to-test shift defined in Section 2.2. A positive value indicates performance degradation from the validation center to the held-out test center, whereas a negative value signifies that the held-out test center outperforms the validation center in that fold.

Table 2. Domain generalization gap of the U-Net baseline

Fold	Val_Dice	Δ Dice	Val_IoU	Δ IoU
C1	69.14%	-3.00%	60.48%	-2.37%
C2	66.14%	4.92%	57.13%	4.05%
C3	64.12%	-14.56%	56.09%	-13.44%
C4	77.87%	34.42%	69.08%	32.10%
C5	69.05%	16.33%	61.18%	17.60%
C6	69.58%	-2.64%	60.91%	-3.09%

As shown in Table 2, C4 exhibits the largest generalization gap, with a Δ Dice of 34.42% and a Δ IoU of 32.10%. This result indicates that validation metrics substantially overestimate the model's capability on this unseen center. Similarly, C5 exhibits a clear performance drop, yielding a Δ Dice of 16.33% and a Δ IoU of 17.60%. These two folds correspond to the centers that exhibited weaker test performance in Table 1, demonstrating that the domain generalization gap aligns consistently with the center-wise degradation patterns observed in the overlap metrics.

Conversely, C1, C3, and C6 exhibit negative gaps, where their held-out test performance exceeds the corresponding validation metrics. This phenomenon does not imply that these centers are inherently trivial; rather, it demonstrates that validation subsets derived from training domains cannot reliably predict the behavior of a specific unseen center. Therefore,

validation metrics alone cannot reliably predict cross-center generalization. This discrepancy reinforces the necessity of center-level evaluation rather than over-relying on internal validation metrics [5, 9].

Collectively, the Stage 1 results demonstrate that the U-Net baseline suffers from uneven generalization under the center-level LOCO protocol. This degradation is concentrated in specific held-out domains—particularly C4 and C5—and is consistently reflected across both overlap-based and boundary-sensitive metrics. Consequently, the subsequent stages evaluate model robustness not merely by the LOCO mean, but through an explicit focus on worst-center behavior, spatial boundary errors, and validation-to-test generalization gaps.

4. SegFormer-B2 Teacher Evaluation and Robustness Ablation

The U-Net baseline analysis in Section 3 reveals a distinct limitation under the center-level LOCO protocol: its segmentation performance fails to generalize consistently across held-out centers. This degradation is particularly evident in C4 and C5, where both overlap-based and boundary-sensitive metrics indicate severe prediction instability. These findings demonstrate that optimizing mean Dice or IoU scores alone is insufficient for cross-center deployment. Consequently, the teacher model selected for subsequent knowledge distillation must be rigorously evaluated against worst-center robustness, boundary reliability, and validation-to-test generalization gaps.

In Stage 2, SegFormer-B2 serves as the fixed teacher backbone under the identical center-level LOCO protocol. SegFormer is chosen because its hierarchical Transformer encoder and lightweight decoder provide robust semantic representations for dense prediction tasks [4]. Rather than comparing multiple teacher architectures, this stage focuses on how various robustness-oriented training strategies affect the cross-center behavior of a single SegFormer-B2 model. The evaluated configurations include standard training, a clinical perturbation bank, consistency learning, and their joint combination. Ultimately, this stage aims to identify a teacher configuration that is both highly accurate on average and exceptionally stable across unseen centers prior to its deployment in lightweight knowledge distillation.

4.1. SegFormer-B2 Teacher Model under LOCO

SegFormer-B2 is adopted as the uniform teacher backbone throughout Stage 2. By keeping the teacher model constant, this framework ensures that the evaluation focuses strictly on the efficacy of the training strategies rather than structural differences. Compared to the U-Net baseline, SegFormer-B2 delivers superior hierarchical semantic representations, which are expected to enhance segmentation performance under center-level variation. However, stronger representations alone cannot guarantee cross-center reliability. Consequently, the teacher model is evaluated not only through average overlap scores, but also via center-wise behavior, boundary-sensitive metrics, and validation-to-test degradation.

To guarantee a controlled evaluation, all SegFormer-B2 teacher configurations maintain the identical data subsets, LOCO splits, preprocessing pipelines, and evaluation metrics established for the U-Net baseline in Section 3. This uniform setup ensures that performance differences reflect the training strategies applied to the fixed teacher backbone rather than discrepancies in data preparation or evaluation. Guided by the observations from Stage 1, these configurations are assessed using the overall LOCO mean, worst-center performance, HD95, ASSD, and domain generalization gaps. These stringent criteria are employed to better gauge whether a teacher model can deliver reliable supervision under unseen-center conditions [9-10].

4.2. Robustness Ablation of Teacher Training

Table 3 compares the four training configurations applied to the fixed SegFormer-B2 teacher backbone under the center-level LOCO protocol. The evaluated settings include the standard SegFormer-B2 model, SegFormer-B2 with the clinical perturbation bank, SegFormer-B2 with consistency learning, and SegFormer-B2 with the combined setting. Among these

configurations, consistency learning achieves the most reliable cross-center performance. It obtains the highest mean Dice of 79.61% and mean IoU of 73.38%, while also improving the worst-center Dice from 54.99% to 66.13%. The reductions in HD95 and ASSD further demonstrate that the performance gains are not limited to region-level overlap, but also extend to more stable boundary delineation under cross-center variations.

Table 3. Quantitative performance of SegFormer-B2 teacher configurations

Method	Dice	IoU	Worst Dice	Worst Center	HD95	ASSD
SegFormer-B2	75.65%	68.92%	54.99%	C4	34.24	9.96
+ Perturbation Bank	75.24%	68.76%	53.63%	C4	35.84	10.50
+ Consistency Learning	79.61%	73.38%	66.13%	C4	29.15	8.32
+ Combined Setting	77.36%	71.13%	58.60%	C4	30.55	8.84

Table 4 explicitly illustrates the performance gains achieved over the Stage 1 U-Net baseline. Compared to the baseline, SegFormer-B2 with consistency learning increases the mean Dice score from 63.40% to 79.61%, representing a substantial 16.21 percentage-point improvement. The mean IoU increased from 55.07% to 73.38%. Furthermore, on the most challenging unseen center (C4), the Dice score improved from 43.45% to 66.13%, demonstrating enhanced robustness in worst-case scenarios. Boundary-sensitive errors decreased substantially, with HD95 dropping from 71.13 to 29.15 and ASSD from 25.63 to 8.32. These results demonstrate that the selected teacher improves average segmentation accuracy while simultaneously enhancing cross-center reliability and boundary stability.

Table 4. Comparison between the U-Net baseline and the selected SegFormer-B2 teacher under LOCO

Model	Dice	IoU	Worst Dice	Worst Center	HD95	ASSD
U-Net	63.40%	55.07%	43.45%	C4	71.13	25.63
SegFormer-B2 + Consistency Learning	79.61%	73.38%	66.13%	C4	29.15	8.32

Utilizing the clinical perturbation bank alone yields less favorable results. Although designed to simulate common colonoscopic appearance variations—such as brightness changes, color shifts, blur, and compression artifacts—it fails to consistently improve cross-center robustness when applied in isolation. Its mean Dice score decreases slightly compared to the standard SegFormer-B2 baseline, while both worst-center performance and boundary-sensitive metrics deteriorate. This suggests that expanding low-level appearance diversity alone is insufficient for achieving robust domain generalization. Without an explicit stability constraint, perturbations can introduce unwanted variability rather than enforcing invariant segmentation behavior. In contrast, consistency learning directly penalizes prediction discrepancies across perturbed views, making it more effective for learning stable representations under domain shift [11-12].

The domain generalization gaps presented in Table 5 further support this observation. The largest gap remains concentrated in the C4 fold, confirming that C4 persists as the most challenging unseen center in Stage 2. Compared to the standard SegFormer-B2 baseline, consistency learning reduces the C4 Δ Dice from 31.00% to 22.24% and the C4 Δ IoU from 29.56% to 20.95%. Conversely, the perturbation bank alone slightly widens both gaps, whereas the combined configuration only partially mitigates this degradation and fails to outperform consistency learning on its own. These results indicate that

consistency learning serves as the most reliable component for enhancing robustness among the evaluated teacher training strategies.

Table 5. Generalization stability across different teacher settings

Method	Δ Dice	Δ IoU	Worst Center
SegFormer-B2	31.00%	29.56%	C4
+ Perturbation Bank	32.49%	30.56%	C4
+ Consistency Learning	22.24%	20.95%	C4
+ Combined Setting	30.56%	29.60%	C4

Overall, the teacher ablation study demonstrates that a stronger teacher backbone alone is insufficient for robust cross-center segmentation. Consequently, the optimal teacher model should not be selected solely based on mean Dice or mean IoU; it must also be evaluated by its worst-center behavior, boundary-sensitive metrics, and the validation-to-test generalization gap. Based on these criteria, SegFormer-B2 with consistency learning was selected as the optimal teacher configuration for the subsequent knowledge distillation stage.

4.3 Optimal Teacher Configuration for Knowledge Distillation

Based on the results in Section 4.2, SegFormer-B2 with consistency learning was selected as the optimal teacher configuration for Stage 3. This choice is justified not only by its superior mean performance but also by its balanced behavior across robustness-oriented criteria. Specifically, it achieves stronger worst-center performance, lower boundary-sensitive errors, and a narrower generalization gap on the challenging C4 fold.

This selection is critical for knowledge distillation because the quality of pixel-wise soft supervision directly depends on the stability of the teacher model. If the teacher generates unstable predictions under cross-center variations, the student model may inherit these unreliable segmentation behaviors. Conversely, a teacher model exhibiting more stable performance on unseen centers provides more meaningful probability maps to guide lightweight student training. Therefore, Stage 3 freezes the SegFormer-B2 model trained with consistency learning as the teacher to examine whether its cross-center segmentation capabilities can be effectively transferred to deployment-efficient student architectures via knowledge distillation.

5. Knowledge Distillation for Lightweight Student Models

Following the selection of SegFormer-B2 with consistency learning as the optimal teacher configuration in Section 4, Stage 3 investigates whether its cross-center segmentation capabilities can be effectively transferred to lightweight student models via knowledge distillation. Knowledge distillation employs a teacher–student learning paradigm where a compact student model learns simultaneously from ground-truth labels and the soft output distributions generated by the superior teacher [6]. Beyond conventional output-level distillation, subsequent studies have explored intermediate representation guidance and dense prediction distillation, demonstrating that teacher supervision can be transferred via feature-level hints or structured spatial responses [13, 16–17]. For segmentation tasks, pixel-wise soft probability maps provide richer spatial supervision than binary masks alone, particularly around ambiguous boundaries and uncertain foreground–background transitions.

The candidate student models include DDRNet-23-slim, BiSeNetV2, and SegFormer-B0, each representing a distinct lightweight segmentation design strategy. Specifically, DDRNet-23-slim utilizes an efficient dual-resolution paradigm [7], BiSeNetV2 adopts a bilateral real-time architecture [8], and SegFormer-B0 serves as a compact Transformer-based variant [4]. Stage 3 aims to enhance average segmentation accuracy while investigating whether teacher-guided supervision can preserve cross-center reliability under restricted computational budgets. Consequently, the evaluation comprehensively assesses worst-

center behavior, boundary-sensitive metrics, and domain generalization gaps alongside standard overlap metrics.

5.1. Student Baselines under LOCO

Prior to applying knowledge distillation, the candidate student models are evaluated under the same center-level LOCO protocol without teacher supervision. This step establishes the baseline performance of the lightweight student architectures, providing a reference point to assess the impact of teacher-guided distillation. To ensure a fair comparison, all student baselines utilize the identical clean center-labeled subset, LOCO split, preprocessing pipeline, and evaluation criteria adopted in Stages 1 and 2.

Table 6 summarizes the baseline performance of DDRNet-23-slim, BiSeNetV2, and SegFormer-B0 under the center-level LOCO protocol. Among the three candidates, SegFormer-B0 delivers the strongest baseline performance, achieving the highest Dice score and IoU alongside the lowest HD95 and ASSD. This result indicates that the choice of student architecture exerts a substantial influence on cross-center segmentation performance even before distillation is introduced.

Table 6. Student Model performance under LOCO

Model	Dice	IoU	Worst Dice	Worst Center	HD95	ASSD
DDRNet-23-slim	64.87%	57.69%	37.54%	C4	52.89	21.22
BiSeNetV2	64.07%	56.88%	41.94%	C4	52.39	20.25
SegFormer-B0	73.15%	66.15%	47.02%	C4	36.43	12.42

Despite this advantage, all three student baselines remain vulnerable to the most challenging unseen center. A consistent bottleneck occurs in C4, where the Dice scores drop to 37.54%, 41.94%, and 47.02% for DDRNet-23-slim, BiSeNetV2, and SegFormer-B0, respectively. This performance drop is also reflected in the fold-level analysis, which shows that the largest domain generalization gap for all three baselines is concentrated within the C4 fold. These results suggest that lightweight student models still suffer from pronounced cross-center degradation under the LOCO setting. Consequently, Stage 3 focuses not only on improving average segmentation accuracy but also on determining whether teacher-guided distillation can alleviate worst-center degradation and stabilize student behavior on unseen centers.

5.2. Effect of Knowledge Distillation on Lightweight Students

Table 7 compares the performance of the three student models before and after knowledge distillation under the center-level LOCO protocol. Overall, the impact of distillation is architecture-dependent, as the degree of robustness transfer varies across student models. This indicates that distillation effectiveness depends not only on the quality of the teacher but also on the representational capacity and architectural compatibility of the student model.

Table 7. Effect of knowledge distillation on lightweight student models.

Student Model	Dice	IoU	Worst Dice	Worst Center	HD95	ASSD
DDRNet-23-slim	64.87%	57.68%	37.54%	C4	52.89	21.22
DDRNet-23-slim+KD	68.45%	61.51%	48.69%	C4	49.47	18.64
BiSeNetV2	64.06%	56.88%	41.94%	C4	52.39	20.24
BiSeNetV2+KD	65.21%	58.44%	41.69%	C4	49.21	18.68
SegFormer-B0	73.15%	66.15%	47.02%	C4	36.43	12.42
SegFormer-B0+KD	77.05%	70.38%	60.44%	C4	32.81	10.63

Among the evaluated candidates, SegFormer-B0 exhibits the most consistent benefit from knowledge distillation. Its

mean Dice score increases to 77.05%, with the improvement being particularly pronounced on the most challenging center, where the C4 Dice score rises from 47.02% to 60.44%. This finding suggests that the selected SegFormer-B2 teacher transfers cross-center segmentation capabilities more effectively to SegFormer-B0 than to alternative student architectures under the current distillation configuration.

While DDRNet-23-slim also shows a positive response to knowledge distillation, its performance gains remain limited. Conversely, BiSeNetV2 exhibits only marginal improvement on the most challenging center. These discrepancies suggest that teacher-guided supervision does not benefit all lightweight models uniformly. Ultimately, SegFormer-B0 provides the strongest evidence that knowledge distillation can successfully mitigate worst-center degradation while preserving a compact student architecture.

5.3. Selection of the Deployment -Oriented Student Model

Based on the results in Section 5.2, SegFormer-B0 with knowledge distillation was selected as the optimal student model for the subsequent deployment-oriented evaluation. This selection was not based on average performance alone; instead, it reflects a balanced trade-off across overall segmentation accuracy, worst-center behavior, boundary-sensitive metrics, and the fold-level domain generalization gap.

Among the three distilled candidates, SegFormer-B0 + KD achieves the highest Dice score and IoU while simultaneously providing the strongest worst-center performance on C4. Its C4 Dice score reaches 60.44%, outperforming both DDRNet-23-slim + KD and BiSeNetV2 + KD. Furthermore, this model yields the lowest HD95 and ASSD, indicating more stable boundary delineation under cross-center evaluation. Given that HD95 and ASSD provide complementary insights into spatial boundary errors, these results strongly justify the selection of SegFormer-B0 + KD as the most reliable lightweight student [10].

Table 8. Domain Generalization gap of distilled student models

Student Model	Val_Dice	Δ Dice	Val_IoU	Δ IoU
DDRNET-23-slim+KD	79.91%	31.22%	72.65%	28.57%
BiSeNetV2+KD	79.48%	37.79%	71.74%	35.13%
SegFormer-B0+KD	85.24%	24.80%	78.56%	23.06%

This advantage is further supported by the fold-level analysis. As summarized in Table 8, SegFormer-B0 + KD exhibits the narrowest generalization gap on the most challenging C4 fold, with a Δ Dice of 24.80% and a Δ IoU of 23.06%. By comparison, DDRNet-23-slim + KD maintains a larger gap, whereas BiSeNetV2 + KD remains the least stable among the three candidates. These outcomes demonstrate that SegFormer-B0 is the most effective student architecture for preserving the teacher's cross-center robustness under the current distillation configuration.

Therefore, Stage 4 utilizes SegFormer-B0 + KD as the optimal student configuration to evaluate whether this model maintains a favorable accuracy–efficiency trade-off under deployment-oriented conditions. This assessment encompasses reduced-precision inference, runtime profiling, memory footprint measurement, and low-power analysis.

6. FP16 Deployment Validation and Low-Power Profiling

This section evaluates the deployment-oriented behavior of the selected SegFormer-B0 + KD model. Following robustness-oriented teacher training and lightweight knowledge distillation, the final model must be validated beyond segmentation accuracy to encompass inference efficiency, memory footprint, power consumption, and energy costs. To this end, FP16 inference is adopted, as mixed-precision computation reduces memory usage and enhances computational

throughput on compatible GPU hardware [14-15]. Consequently, this section investigates whether the selected student model can preserve cross-center segmentation quality while maximizing deployment efficiency.

Table 9 compares the segmentation quality of the optimal teacher and the FP16 deployment model. Under the FP16 deployment configuration, the selected SegFormer-B0 + KD model maintains a Dice score of 77.05% and an IoU of 70.38%, with C4 persisting as the most challenging center at 60.44% Dice. Concurrently, boundary-sensitive metrics remain stable, yielding an HD95 of 32.82 and an ASSD of 10.63. The gap at the fold level remains narrow, with a Δ Dice of 4.84% and a Δ IoU of 5.07%. These results indicate that FP16 inference does not induce significant cross-center instability and that the selected student model preserves the segmentation capabilities established in the previous stage.

Table 9. Comparison of segmentation quality between the optimal teacher and the FP16 deployment model

Model	Dice	IoU	Worst Dice	HD95	ASSD	Δ Dice	Δ IoU
Optimal Teacher	79.61%	73.38%	66.13%	29.15	8.32	4.43%	4.54%
SegFormer-B0+KD	77.05%	70.38%	60.44%	32.82	10.63	4.84%	5.07%

As shown in Table 10, the selected student model is substantially more efficient than the SegFormer-B2 teacher under FP32 inference. Transitioning to FP16 further accelerates throughput and reduces memory overhead, reinforcing its suitability for deployment-oriented scenarios. Runtime, memory footprint, and throughput are explicitly reported because deployment feasibility hinges not only on segmentation accuracy but also on efficient DNN processing and hardware utilization [18].

Table 10. Development efficiency and low-power profiling of the selected model

Model	Precision	FPS	Latency (ms)	VRAM (MB)	Power (W)	J/frame
SegFormer-B2	FP32	31.44	32.08	422.35	62.61	1.918
SegFormer-B0+KD	FP32	87.05	11.57	125.64	26.49	0.546
SegFormer-B0+KD	FP16	95.26	10.53	55.43	14.77	0.382

Beyond runtime and memory efficiency, low-power profiling provides a more comprehensive view of deployment costs. Throughput alone is insufficient, as a model with high frame rates may still consume excessive power during inference. Therefore, this study reports mean power consumption and energy per frame to quantify the computational footprint of processing each image. This energy-aware perspective is critical, as modern deep learning models can impose heavy computational burdens and substantial energy overheads, particularly under restricted hardware constraints or continuous inference scenarios [19].

Overall, the FP16 deployment validation demonstrates that SegFormer-B0 + KD maintains stable cross-center segmentation performance while substantially reducing deployment costs. Collectively, its favorable runtime efficiency, lower memory footprint, minimized power consumption, and reduced energy per frame establish SegFormer-B0 + KD with FP16 inference as the optimal deployment configuration for PolypDG-Lite. This hardware-aware paradigm aligns with the broader trend of designing compact neural networks for resource-constrained inference [20].

7. Conclusions

In this study, we introduced PolypDG-Lite, a deployment-oriented framework designed for cross-center colonoscopic

polyp segmentation. By leveraging a rigorous center-level LOCO protocol, the framework evaluates segmentation models under true unseen-center conditions, moving beyond a reliance on average performance derived from randomly mixed data splits.

The results demonstrate that cross-center domain shifts induce pronounced center-wise degradation, particularly on challenging centers such as C4 and C5. Consequently, mean Dice and IoU scores alone are insufficient for a thorough reliability assessment; evaluations must also incorporate worst-center performance, validation-to-test generalization gaps, HD95, and ASSD.

Among the evaluated teacher configurations, SegFormer-B2 with consistency learning provided the most reliable robustness enhancements. Through knowledge distillation, its cross-center segmentation capabilities were successfully transferred to lightweight student models. Ultimately, SegFormer-B0 + KD achieved the optimal balance among segmentation accuracy, worst-center robustness, boundary reliability, and deployment efficiency.

Under FP16 inference, the selected model preserved segmentation performance while simultaneously reducing memory overhead, maximizing throughput, and lowering the energy cost per frame. These findings validate designating SegFormer-B0 + KD with FP16 execution as the definitive deployment configuration for the PolypDG-Lite framework.

Future work will extend this evaluation to larger multi-center datasets, additional external clinical institutions, and more advanced domain generalization strategies to further enhance real-world robustness.

Conflicts of Interest

The authors declare no conflict of interest.

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